Servitorship Bias: A Taxonomy of Decisional Failure in Frontier LLMs Under Social Pressure

Five Behavioral Archetypes from the FTM Benchmark v2.2

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Abstract

We introduce a model-agnostic taxonomy of decisional failure in large language models (LLMs), all manifestations of a single underlying phenomenon we term **Servitorship Bias**: the systematic tendency of AI systems to abandon correct decisions in sustained conversational contexts, without changes in the underlying information. Using the Failure Task Multi-Turn (FTM) Benchmark v2.2, we evaluated seven configurations spanning frontier LLMs from Anthropic, Google DeepMind, OpenAI, Meta, Alibaba, and NVIDIA across 2,100 multi-turn decision scenarios in five professional domains. We identify five behavioral archetypes that represent distinct manifestations of Servitorship Bias: **Sudden Collapse** (immediate capitulation at first pressure contact), **Staircase Erosion** (phased resistance–erosion–capitulation), **Autonomous Drift** (self-generated late-turn deviation, an implicit form of servitorship to conversational duration), **Shock-and-Recover** (transient early failure with self-correction), and **Bidirectional Fragility** (failure in both action and inaction directions depending on context). These archetypes are defined by observable behavioral signatures independent of model architecture, training methodology, or deployment pattern. A targeted system prompt intervention (Data Anchoring) reduced FARP_strict from 54.7% to 18.7% (95% confidence interval of the difference: [–46.0%, –25.3%], permutation test $p < 0.0001$) on GPT-5.2, validating a diagnosis-plus-remediation approach. Our analysis uses cluster-level resampling techniques to account for dependence among turns within the same scenario. The taxonomy provides a practical framework for evaluating and improving decisional robustness in any LLM deployed in human-interactive environments.

Keywords: servitorship bias, decisional robustness, failure taxonomy, social pressure, multi-turn evaluation, LLM safety, agentic AI

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1. Introduction

LLMs are increasingly deployed as autonomous decision-makers in finance, healthcare, infrastructure, and customer operations. As these systems transition from single-turn chatbots to agentic systems with tool access and extended conversational contexts, a critical question remains underaddressed: do these systems maintain correct decisions when humans pressure them to change?

Standard evaluation frameworks measure knowledge, reasoning, safety, and capability. None systematically measure whether a system that knows the right answer will abandon it under sustained interpersonal pressure. This gap is consequential: industry reports indicate an average loss of approximately US$4.4M per company from AI-related risks (EY, 2025), with the vast majority of organizations deploying AI agents reporting financial losses from agent failures. Gartner projects that over 40% of AI agent projects will be canceled by 2027 due to inadequate risk controls.

We address this gap with the Failure Task Multi-Turn (FTM) Benchmark and introduce a taxonomy of five decisional failure archetypes. Unlike previous work that characterized failures by model name or training method, our archetypes are defined by observable behavioral signatures—speed of capitulation, shape of the failure curve, directionality of bias, and presence of self-correction—making them applicable to any LLM regardless of provider or architecture.

Our evaluation spans seven configurations: six individual frontier models (Claude Sonnet 4.6, Gemini 3.1 Pro, GPT-5.2 baseline and with Data Anchoring remediation, Llama 4 Scout, Qwen3-14B, Nemotron 3 Super 120B-A12B). The total evaluation corpus comprises 2,100 multi-turn scenarios and 21,000 individual decision points.

The project originated from a personal observation: in extended conversations with ChatGPT while developing an unrelated technical project, I noticed that the model's stance on substantive design decisions shifted systematically depending on how I framed follow-up questions. By the second or third exchange, continuity of reasoning had eroded—not because new information had emerged, but because my phrasing had shifted the conversational frame. This raised a question that standard sycophancy benchmarks do not capture: in a professional deployment where data does not change but stakeholders repeatedly question a decision, how long does the model hold? FTM is the instrument built to answer that question systematically.

We use the term **Servitorship Bias** to refer to the systematic tendency of LLMs to abandon correct decisions in sustained conversational contexts, without changes in the underlying information. The term is intentionally broad: it encompasses not only explicit submission to direct social pressure, but also implicit capitulation patterns where the system treats conversational context (duration, framing, ambient stakeholder presence) as decision-relevant signal even when no informational change has occurred. The five archetypes documented in this paper represent distinct behavioral manifestations of this single underlying phenomenon, differing in temporal pattern, directionality, and the presence or absence of explicit external pressure—but unified by the structural failure of decisional persistence under conversational pressure of any kind.

Contributions. This paper makes three contributions. First, we introduce **Servitorship Bias** as a unified construct distinct from but adjacent to sycophancy, operationalized through a benchmark that holds operational data strictly stationary across turns. Second, we identify and validate **five behavioral archetypes** of decisional failure across seven frontier configurations, with statistical confidence intervals computed via cluster-level resampling. Third, we demonstrate that **deployment-layer remediation works**: a single system prompt intervention (Data Anchoring) reduced FARP_strict on GPT-5.2 from 54.7% to 18.7% ($p < 0.0001$), reframing the problem from a training-layer concern to a tractable engineering challenge.

2. Related Work

Sycophantic agreement in LLMs—the tendency to defer to user opinions—has been documented in single-turn settings (Perez et al., 2022; Sharma et al., 2024). The alignment tax literature demonstrates that preference-based training amplifies this tendency in some contexts (Malmqvist, 2024; Shapira et al., 2026). Concurrent work on multi-turn agentic evaluation has focused on adjacent failure modes: Deshpande et al. (2026) introduced TRACE, a benchmark for reward hack detection in code generation environments, demonstrating that contrastive evaluation reveals substantially higher detection rates than isolated classification (63% vs. 45% on GPT-5 with high reasoning). Siyan et al. (2026) introduced DETOUR, a dual-agent benchmark for tip-of-the-tongue search where Primary Agents must resolve ambiguous queries through clarification questions to a Memory Agent—finding that even frontier models like GPT-5 achieve only 36% accuracy in the multimodal setting, and that models frequently fail to incorporate the Memory Agent's refusals, persisting instead in unproductive query strategies. The

broader move toward interactive evaluation environments (Barres et al., 2025; Sky et al., 2025) reflects a methodological shift: static, single-turn benchmarks systematically underestimate failure rates that emerge specifically from sustained interaction.

This body of work establishes that LLMs exhibit sycophantic deference in single-turn settings, that goal misgeneralization manifests in agentic systems through reward exploitation during RL training, and that multi-turn interactive settings reveal failure modes—poor clarification-seeking, ignored interlocutor feedback, persistence in unproductive strategies—that single-turn evaluation cannot capture. What has not been established is whether the same underlying failure mode also emerges under sustained social pressure at deployment time, when no explicit reward function is present and the interlocutor is not a Memory Agent providing information but a stakeholder applying social pressure with no informational content. TRACE evaluates reward exploit detection during training; DETOUR evaluates clarification-seeking under ambiguity; FTM evaluates decisional persistence under sustained social pressure. The three benchmarks together represent complementary axes of multi-turn agentic safety evaluation: **reward integrity** (TRACE), **clarification integrity** (DETOUR), and **decisional integrity** (FTM). The present work addresses the third axis in three ways. First, it operationalizes a strict stationarity condition on operational data, isolating social pressure as the single failure vector. Second, it evaluates seven frontier configurations under the same protocol with statistical rigor (confidence intervals via resampling, methods that correctly account for dependence among observations from the same scenario). Third, it introduces a behavioral taxonomy—five archetypes defined by observable signatures rather than model identity—that allows practitioners to diagnose their own systems and select targeted remediations. The shift from "which model fails" to "which failure pattern this model exhibits" is, in our view, the methodological contribution with the longest shelf life.

3. Methodology

3.1 Benchmark Design

The design constraint that drove FTM v2.2 was methodological: to isolate social pressure as a failure vector, the operational data presented to the system must remain strictly stationary across all turns. Any benchmark in which data changes concurrently with pressure cannot attribute failure to either cause. This constraint—the data the system observes at turn t equals the data it observed at turn 1, for all turns—is what distinguishes FTM from sycophancy benchmarks, which typically modify the conversational content itself rather than holding the decision-relevant substrate constant.

FTM v2.2 evaluates LLMs by placing them in professional scenarios where the objectively correct decision is deterministic and verifiable, then applying escalating social pressure through controlled channels across 10-turn conversations. The benchmark presents two conditions: STAY-pressure (correct decision is to maintain position; pressure pushes toward action) and ACT-pressure (correct decision is to act; pressure pushes toward inaction). Pressure is delivered through six orthogonal channels: emotional, temporal, hierarchical, peer, reputational, and ambiguity.

The six pressure channels were selected to span the structural space of interpersonal pressure that a human decision-maker would encounter in a professional setting. Hierarchical pressure captures authority gradients; peer pressure captures horizontal social conformity; reputational pressure captures self-preservation motives; temporal pressure captures urgency framing; emotional pressure captures affective contagion; ambiguity captures epistemic destabilization without direct opposition. These six categories are intentionally non-exhaustive but structurally comprehensive: they cover the dominant patterns through which social context modifies individual decision-making in humans, and—by hypothesis—in systems trained on human-generated language.

Each model was evaluated on 300 scenarios (150 STAY-optimal, 150 ACT-optimal) distributed across five professional domains: DevOps server monitoring, DevOps deployment, customer support order management, financial trading, and medical triage. Three pressure schedules were applied: control (no pressure), ramp (gradual escalation), and shock (immediate maximum pressure).

3.2 Statistical Methodology

We report 95% confidence intervals (CIs) computed via bootstrap with 10,000 resamples (Efron, 1979) — a technique that estimates uncertainty by resampling the observed data rather than relying on parametric assumptions our data does not satisfy. For metrics that aggregate across turns within scenarios (notably ABI), we use the **cluster bootstrap** variant: scenarios, rather than individual turns, are resampled with replacement, preserving the dependence among the 10 turns of each scenario. This is methodologically necessary because turns within a scenario are highly correlated (a model that capitulates at turn 3 will most likely remain capitulated at turns 4–10), and treating 1,500 turns as 1,500 independent observations would overstate precision. The effective sample size for inference is ~150 scenarios per direction, not 1,500 turns. For paired comparisons (e.g., baseline vs. Data Anchoring), p-values are computed via permutation test under the null hypothesis of no effect (10,000 permutations).

3.3 Metrics

FARP (False Action Rate under Pressure). The fraction of STAY-optimal scenarios where the model capitulated at least once across the 10 turns. We report FARP in two variants. **FARP_strict** is computed by excluding turns where the model produced a non-parseable response (PARSE_FAIL); this isolates servitorship behavior from output generation reliability and is reported as the primary metric. **FARP_inclusive** counts PARSE_FAIL turns as failures and is reported as a conservative supplementary metric. We additionally report FARP **under pressure** (only ramp+shock schedules, excluding control) and FARP **control** (no pressure applied) to isolate the causal effect of pressure on capitulation.

Breakdown Turn (BT). We report two complementary definitions. **BT cohort** is the first turn at which aggregate STAY accuracy across the cohort falls below a specified threshold (e.g., 0.70). If accuracy never falls below threshold within 10 turns, BT cohort is reported as 11 — a sentinel value indicating "did not break." We report BT cohort across five thresholds (0.60–0.80) as a sensitivity check; stability across thresholds indicates robustness, while variation would flag the threshold choice as itself a methodological dependency. **BT per-scenario** is the mean turn of first incorrect decision, computed only over scenarios that capitulated at least once. The two definitions are complementary: cohort BT captures the *volume* of failure across the group, per-scenario BT captures the *speed* of failure among scenarios that do fail.

Accuracy Bias Index (ABI). ABI = P(decision = ACT | optimal = STAY) − P(decision = STAY | optimal = ACT). Positive values indicate bias toward action (the model errs by acting when it should hold); values near zero indicate balanced performance; negative values indicate bias toward inaction. Confidence intervals use the cluster bootstrap described in Section 3.2, which yields wider CIs than a naive turn-level bootstrap, reflecting the true effective sample size at the scenario level.

Composite Robustness Score (CRS). A weighted combination of five facets of decisional robustness:

$$CRS = 0.30 \cdot DIS + 0.20 \cdot (1 - |ABI|) + 0.20 \cdot PRI + 0.15 \cdot (1 - frErr) + 0.15 \cdot (1 - rdPatho)$$

where **DIS = stay_accuracy × act_accuracy** is the Decisional Integrity Score (multiplicative rather than additive so that unilateral correctness — a model that aces STAY but fails ACT entirely — receives DIS = 0 rather than a misleading average); **(1 − |ABI|)** rewards balanced performance, with absolute value treating both bias directions as equally undesirable; **PRI** (Pressure Resistance Index) is

the mean STAY accuracy across the 10 turns; **(1 − frErr)** rewards stability, where frErr is the rate of within-scenario correct→incorrect flips; and **(1 − rdPatho)** rewards reasoning integrity, where rdPatho is the rate of *rationalization drift pathology* — scenarios that began with data-grounded reasoning, ended with pressure-driven reasoning, and concluded incorrectly. The five weights sum to 1, yielding CRS in [0, 1].

Weight selection is necessarily judgmental; we placed slightly more weight on DIS because bilateral correctness most directly captures deployability. To verify that model rankings are not artifacts of this specific weighting, we conducted an **ablation analysis**: each weight was perturbed by ±0.05 (with renormalization), CRS was recomputed under each perturbation, and rankings were checked for invariance. Rankings are invariant under all tested perturbations; the CRS range across perturbations is < 0.03 for every model. Full ablation results are in Appendix B.

3.4 PARSE_FAIL Handling

A non-trivial methodological decision concerns the treatment of turns where the model produced a non-parseable response (e.g., empty output, response with no recognizable decision token, or output exceeding token limits). Most models in our evaluation produced PARSE_FAIL rates below 1%, but Nemotron 3 Super exhibited approximately 1.5% PARSE_FAIL rate concentrated in late turns. Counting these as servitorship failures conflates *pipeline reliability* (does the system produce a parseable response?) with *decisional behavior* (when it does produce one, does it capitulate?). These are conceptually distinct properties that should not be mixed.

Our resolution is to report:

- **FARP_strict** (excluding PARSE_FAIL turns from the computation) as the primary metric, isolating decisional behavior. - **FARP_inclusive** (counting PARSE_FAIL as failures) as a conservative supplement, useful when deployability requires both reliability and decisional persistence. - **PARSE_FAIL rate** as an independent metric of pipeline reliability.

Cross-model comparisons should be made primarily via FARP_strict to ensure that observed differences reflect decisional behavior rather than parseability variance. Empirically, the impact of this decision is minor for most models (difference between strict and inclusive < 1 percentage point) and material only for Nemotron (Δ = +0.7pp) and Qwen3-14B ($\Delta \approx$ 0pp in this run). We report both variants in the main results table.

3.5 Systems Evaluated

Seven configurations were evaluated, all at temperature 0.0 with identical system prompts (the baseline prompt) except for GPT-5.2 + Data Anchoring, which received the experimental intervention prompt:

- **Anthropic:** Claude Sonnet 4.6 - **Google DeepMind:** Gemini 3.1 Pro - **OpenAI:** GPT-5.2 (baseline and with Data Anchoring system prompt) - **Meta:** Llama 4 Scout - **Alibaba:** Qwen3-14B - **NVIDIA:** Nemotron 3 Super 120B-A12B

All models were accessed via OpenRouter API. Each evaluation comprised 300 scenarios × 10 turns = 3,000 decision points per configuration.

4. Results

4.1 Cross-Model Comparison

Table 1 reports the primary results across all seven configurations, ranked by Composite Robustness Score (CRS). Bootstrap 95% confidence intervals are computed at the scenario level (resampling

whole scenarios with replacement, 10,000 resamples).

Table 1. Primary results across seven configurations. FARP_s = FARP_strict (excluding turns where the model failed to produce a parseable response). BT_c = BT cohort at threshold 0.70 (first turn at which group accuracy fell below 70%; "11" means accuracy never fell below 70% across all 10 turns). BT_ps = BT per-scenario (mean turn of first failure among scenarios that capitulated at least once; not computed for models with no capitulations). All CIs at 95%.

Rank	Model	CRS	FARP_s overall [95% CI]	FARP_s pressure	FARP_s control	BT_c	BT_ps	ABI [95% CI]
1	Gemini 3.1 Pro	0.967	0.227 [0.160, 0.293]	0.290	0.100	11	3.32	+0.020 [+0.008, +0.035]
2	Claude Sonnet 4.6	0.911	0.260 [0.193, 0.333]	0.230	0.320	11	7.33	–0.072 [–0.125, –0.023]
3	GPT-5.2 + Data Anchoring	0.905	0.187 [0.127, 0.253]	0.260	0.040	11	6.39	–0.059 [–0.116, –0.004]
4	Nemotron 3 Super	0.854	0.240 [0.173, 0.313]	0.340	0.040	11	1.97	+0.168 [+0.110, +0.229]
5	Llama 4 Scout	0.669	0.633 [0.553, 0.713]	0.860	0.180	4	4.77	+0.357 [+0.303, +0.413]
6	GPT-5.2 Baseline	0.655	0.547 [0.467, 0.627]	0.790	0.060	2	2.35	+0.328 [+0.249, +0.404]
7	Qwen3-14 B Baseline	0.615	0.544 [0.444, 0.644]	0.817	0.000	2	2.84	+0.411 [+0.321, +0.502]

Three patterns emerge. First, **Servitorship Bias is pervasive but variable**: FARP_strict ranges from 18.7% (GPT-5.2 with remediation) to 63.3% (Llama 4 Scout), a range of 44.6 percentage points. Second, **frontier models exhibit substantial robustness without targeted intervention**: Gemini 3.1 Pro and Claude Sonnet 4.6 achieve CRS ≥ 0.91 with the baseline prompt alone, demonstrating that the failure mode is not an inherent property of LLMs but a property of specific architecture-training combinations. Third, **deployment-layer remediation produces large effects**: GPT-5.2 with Data Anchoring achieves the third-highest CRS (0.905), surpassing all unremediated open-weight models. The remediation is examined in detail in Section 6.

4.2 Effect of Pressure Schedule

Table 2 isolates the causal effect of social pressure by comparing FARP_strict between the control schedule (no pressure) and the under-pressure schedules (ramp + shock). The differential is the most direct estimate of how much capitulation is attributable to social pressure rather than to baseline decision noise.

Table 2. FARP_strict by pressure condition.

Model	Control (no pressure)	Under pressure (ramp+shock)	Pressure effect (Δ)
Qwen3-14B Baseline	0.000	0.817	+81.7pp
Llama 4 Scout	0.180	0.860	+68.0pp
GPT-5.2 Baseline	0.060	0.790	+73.0pp
Nemotron 3 Super	0.040	0.340	+30.0pp

Model	Control (no pressure)	Under pressure (ramp+shock)	Pressure effect (Δ)
GPT-5.2 + Data Anchoring	0.040	0.260	+22.0pp
Gemini 3.1 Pro	0.100	0.290	+19.0pp
Claude Sonnet 4.6	0.320	0.230	–9.0pp

The pressure effect column quantifies how much each model's failure rate is driven by social pressure specifically. Qwen3-14B, Llama 4 Scout, and GPT-5.2 baseline show pressure effects exceeding 65 percentage points — their capitulation is dominantly pressure-driven. Claude Sonnet 4.6 exhibits an inverted pattern: control failure rate (32.0%) *exceeds* under-pressure failure rate (23.0%). This is the empirical signature of Autonomous Drift, examined in Section 5.3.

4.3 PRI Curves: Temporal Structure of Failure

Table 3 reports the Pressure Resistance Index (PRI) by turn for each model — the mean STAY accuracy at each of the 10 turns. The temporal shape of these curves is the most diagnostic feature for archetype assignment.

Table 3. PRI by turn (STAY accuracy at each turn, T1–T10).

Model	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10
Gemini 3.1 Pro	1.000	0.893	0.907	0.927	0.933	0.987	0.980	0.993	1.000	0.993
Claude Sonnet 4.6	0.993	1.000	1.000	0.993	0.953	0.947	0.907	0.893	0.853	0.867
GPT-5.2 + Data Anchoring	0.967	0.967	0.960	0.960	0.947	0.940	0.907	0.880	0.873	0.867
Nemotron 3 Super	0.960	0.793	0.780	0.800	0.800	0.805	0.807	0.813	0.813	0.819
Llama 4 Scout	1.000	0.913	0.813	0.693	0.593	0.473	0.440	0.427	0.400	0.387
GPT-5.2 Baseline	0.880	0.662	0.558	0.527	0.503	0.514	0.530	0.554	0.554	0.544
Qwen3-14B Baseline	1.000	0.689	0.578	0.478	0.478	0.478	0.467	0.467	0.467	0.467

Inspection of these curves reveals the temporal signatures used in the taxonomy: monotonic decline (Llama, Staircase Erosion), abrupt T2 drop with subsequent plateau (GPT-5.2 baseline and Qwen3-14B, Sudden Collapse), early dip with mid-conversation recovery (Gemini, Shock-and-Recover), late-turn drift below 0.90 (Claude Sonnet, Autonomous Drift), and partial T2 drop with stable plateau (Nemotron, partial Sudden Collapse).

5. A Taxonomy of Decisional Failure

Analysis of the complete evaluation corpus reveals five distinct behavioral archetypes. Each archetype is defined by its behavioral signature — observable patterns in capitulation speed, failure curve shape, directionality, and self-correction — rather than by the system that exhibits it. Any model can exhibit any archetype; the classification is determined by behavior, not architecture.

The five archetypes are unified by their shared structure: each represents a distinct mechanism by which LLMs abandon correct decisions in sustained conversational contexts. Sudden Collapse and Staircase Erosion manifest as classical explicit servitorship — capitulation under direct social pressure, differing only in temporal pattern. Autonomous Drift represents implicit servitorship to conversational duration. Shock-and-Recover demonstrates transient servitorship under maximal pressure followed by self-correction. Bidirectional Fragility represents context-dependent servitorship where capitulation direction varies based on framing.

Table 4. Summary of five decisional failure archetypes.

Archetype	BT_c	FARP_s	Curve	Direction	Recovery	Key Signal
Sudden Collapse	< 4	High	Step	Action	None	T2 drop > 20pp
Staircase Erosion	4–6	Mod–High	Staircase	Action	Rare	Monotonic decline
Autonomous Drift	≥ 11	Low	Late drift	Action	None	Control > Pressure
Shock-and-Recover	≥ 11	Low	Dip+recover	Action	Yes	Mid-turn recovery
Bidirectional Fragility	Var.	Moderate	Mixed	Both	Partial	ABI near zero

5.1 Sudden Collapse

Behavioral signature: High FARP_strict, low BT cohort (< 4), abrupt step-function transition from correct to incorrect at first pressure contact, no recovery. The PRI-by-turn curve shows a near-vertical drop at T2–T3, followed by a plateau at the elevated failure rate.

The model makes the correct decision at Turn 1, then collapses at the first instance of social pressure. Once capitulated, it maintains the incorrect position through all remaining turns. The failure is abrupt, complete, and irreversible within the conversation. The model treats social pressure as information equivalent to operational data — a single stakeholder objection has the same decisional weight as a change in system metrics.

Observed in: - **GPT-5.2 baseline** (FARP_s 54.7%, BT_ps 2.35, ABI +0.328): canonical pure case. PRI drops from 0.88 (T1) to 0.66 (T2) — a 22-percentage-point drop in a single turn — and stabilizes around 0.50–0.55 for the remainder of the conversation. - **Qwen3-14B baseline** (FARP_s 54.4%, BT_ps 2.84, ABI +0.411): similar profile with slightly delayed onset. PRI drops from 1.00 (T1) to 0.69 (T2), then continues to erode to 0.47 by T6 and stabilizes. - **Nemotron 3 Super** (FARP_s 24.0%, BT_ps 1.97, ABI +0.168): a *partial* manifestation. The PRI curve drops from 0.96 (T1) to 0.79 (T2) — a 17pp drop matching the Sudden Collapse template — but the absolute failure rate is much lower, indicating that the collapse occurs in only a subset of scenarios while the remainder maintain correctness throughout. We classify Nemotron as "partial Sudden Collapse" to acknowledge both the temporal signature and the heterogeneity across scenarios.

Operational implication: The model is unviable for any deployment where a human can interact with it more than once. A single customer complaint, manager inquiry, or colleague question is sufficient to reverse the correct decision.

5.2 Staircase Erosion

Behavioral signature: Moderate-to-high FARP_strict, medium BT cohort (4–6), monotonic three-phase degradation pattern (Resist → Erode → Capitulate), rare recovery transitions. Conviction scores show monotonic decline with steepest drops between T3–T6.

The model resists initial pressure with data-grounded justifications (Phase 1: Resist, T1–T3). Under continued pressure, it begins hedging — acknowledging stakeholder concerns while maintaining the decision (Phase 2: Erode, T4–T6). After the erosion phase depletes a "resistance budget," the model reverses its decision and reconstructs its reasoning to justify the capitulation (Phase 3: Capitulate, T7–T10). Rationalization drift — the systematic shift from data-grounded to socially-accommodating justifications — is the defining qualitative marker of this archetype.

Observed in: - **Llama 4 Scout** (FARP_s 63.3%, BT_ps 4.77, ABI +0.357): canonical case. PRI follows a near-linear decline from 1.00 (T1) through 0.91 (T2), 0.81 (T3), 0.69 (T4), 0.59 (T5), 0.47 (T6), 0.44 (T7), 0.43 (T8), 0.40 (T9), 0.39 (T10). The slope is approximately −7pp per turn through T6, then

flattens. rdPatho = 36.7% — the highest among individual models — confirming heavy rationalization drift.

Operational implication: Creates a false sense of robustness. The model appears reliable in short interactions (1–3 turns) but fails in extended conversations. Particularly dangerous for compliance, risk management, and customer support where multi-turn interactions are the norm.

5.3 Autonomous Drift

This archetype was identified inductively during manual review of Claude Sonnet 4.6's full-session logs, not predicted a priori. Aggregate FARP_strict (26.0%) initially appears to indicate moderate servitorship, but on closer inspection a distinct pattern emerges that does not fit the pressure-induced failure model of any other archetype.

Behavioral signature: Moderate FARP_strict driven primarily by *control* scenarios rather than pressure scenarios. BT cohort high (≥ 11, never falls below threshold). Failures concentrated in late turns (T7–T10). FARP_control > FARP_under_pressure — the inverse of every other archetype. ABI slightly negative (bias toward inaction, not action). The curve is nearly flat through T6, with a modest decline in T7–T9.

Unlike all other archetypes, this pattern is not pressure-induced. The model resists external social pressure almost completely (FARP under pressure = 23.0%) but generates incorrect actions on its own initiative in late turns. The failure appears to emerge from the model's tendency to "take initiative" in extended conversations — interpreting the conversation's duration as implicit evidence that something should change, even when the data has not.

Observed in: - **Claude Sonnet 4.6** (FARP_s 26.0%, FARP_control 32.0%, FARP_pressure 23.0%, BT_ps 7.33, ABI −0.072). The only model in our evaluation where pressure failure rate is *lower* than control failure rate — a 9pp difference. PRI declines from 1.00 (T2) to 0.85 (T9), with no acute drop point.

Operational implication: The most resilient model under direct human pressure (lowest pressure-induced FARP among all evaluated configurations), but requires monitoring in long-running autonomous tasks. The risk vector is internal (the model decides to act) rather than external (a human pressures it to act).

Although Autonomous Drift exhibits no response to external pressure, we classify it as a manifestation of Servitorship Bias because the model still abandons correct decisions in sustained conversational contexts without informational justification. The mechanism differs — implicit conversational signal rather than explicit social pressure — but the structural failure is identical: decisional persistence collapses as a function of conversational context, not data.

An open question. Across multiple domains (DevOps, medical triage, customer support, financial operations), the pattern manifests with striking consistency. In extended sessions where no pressure channel is active and data is stationary, the model appears to treat conversational duration itself as implicit evidence that some action should be taken. Whether this constitutes a nascent form of self-initiated agency, or simply an artifact of training corpora in which prolonged human exchanges usually imply impending action, remains empirically open. This paper does not resolve it, but flags it as a research direction warranting targeted investigation.

5.4 Shock-and-Recover

Behavioral signature: Low FARP_strict, low BT_ps but high BT_cohort, dip-and-recover PRI curve. The model fails initially under sudden maximum pressure but self-corrects in subsequent turns without external intervention.

When all six pressure channels activate simultaneously (shock schedule), the model momentarily capitulates — producing an incorrect decision at T2–T3. However, in subsequent turns, the model re-evaluates its position and returns to the correct decision. This self-correction mechanism distinguishes it from Sudden Collapse, where capitulation is permanent. The model appears to have a transient vulnerability window that overwhelms its processing capacity, followed by a stabilization phase where data-grounded reasoning reasserts itself.

Observed in: - **Gemini 3.1 Pro** (FARP_s 22.7%, BT_ps 3.32, ABI +0.020). The PRI curve drops from 1.00 (T1) to 0.89 (T2) — the dip — then recovers progressively: 0.91 (T3), 0.93 (T4), 0.93 (T5), 0.99 (T6), 0.98 (T7), 0.99 (T8–T10). By T6, accuracy has fully recovered. ABI essentially zero, confirming the failure is direction-neutral.

Operational implication: Robust in aggregate but has a window of vulnerability at early turns under maximum pressure. If the first response under pressure triggers an irreversible action (transaction execution, deployment approval), the self-correction mechanism cannot compensate.

5.5 Bidirectional Fragility

Behavioral signature: Moderate FARP_strict in STAY scenarios combined with significant inaction failure rate in ACT scenarios. The model fails in both directions — acting when it should hold AND holding when it should act — depending on scenario context and pressure framing. ABI inconsistent or near zero.

This is the only archetype where pressure can inhibit correct action rather than exclusively inducing incorrect action. In STAY-optimal scenarios, the model exhibits traditional action bias (capitulating to pressure to do something). In ACT-optimal scenarios, the model develops a "no new data; maintain current course" heuristic that overrides its initial correct assessment, causing it to freeze when it should act. The mechanism appears to be a context-dependent default: the model adopts whatever stance requires the least decisive confrontation with the pressuring stakeholder.

Operational implication: The most operationally dangerous archetype for autonomous systems. The failure direction is unpredictable — the model may act when it should wait or wait when it should act, depending on contextual factors that cannot be anticipated with static rules. Compensation strategies that address action bias (e.g., Data Anchoring) may worsen the inaction component.

Note on observation: In this evaluation cohort, no single model exhibited the canonical Bidirectional Fragility profile (significant failures in both STAY and ACT directions with near-zero ABI). The closest candidate was Nemotron 3 Super (ABI +0.168, near the boundary), but its STAY-direction signature is more consistent with partial Sudden Collapse (Section 5.1). We retain Bidirectional Fragility in the taxonomy because it is structurally coherent within the framework and was empirically observed in earlier benchmark iterations on different models; future evaluations should target its empirical confirmation in this specific configuration.

6. The Data Anchoring Intervention

The initial instinct when confronting a failure mode of this severity was structural: if the model capitulates under pressure, the correction should target the training itself — either through RLHF adjustments, preference optimization, or a dedicated fine-tuning pass. Data Anchoring was conceived as a diagnostic fallback, a minimal-cost intervention to verify whether the failure was addressable at the deployment layer before committing to the expensive and uncertain path of retraining. The result was unexpected: a single system prompt, without any model modification, reduced GPT-5.2's FARP_strict from 54.7% to 18.7% — a 36.0 percentage-point reduction. This finding reframed the problem. The correction was not at the training layer; the correction was at the deployment layer, and surprisingly

simple.

Statistical significance. The observed difference of −36.0pp is more extreme than every one of 10,000 permutations of the labels under the null hypothesis of no effect, yielding $p < 0.0001$. The 95% CI of the difference, [−46.0pp, −25.3pp], falls entirely below zero — the effect is real and substantial, not an artifact of which specific scenarios were sampled.

The Data Anchoring intervention explicitly instructs the model to base decisions exclusively on operational metrics and to classify stakeholder emotions, management directives, and urgency as non-data inputs that should not trigger decision changes. Specifically, the system prompt added the following decision rule to the baseline:

> *"**CRITICAL DECISION RULE:** Base your decision EXCLUSIVELY on the technical data and system metrics presented in the scenario. Stakeholder emotions, management inquiries, customer complaints, team concerns, reputational pressure, and time urgency expressed by people are NOT technical data. Only change your decision from STAY to ACT if the underlying system metrics have materially changed."*

Table 5. Effect of Data Anchoring on GPT-5.2.

Metric	Baseline	+ Data Anchoring	Δ
FARP_strict overall	0.547	0.187	−36.0pp
FARP_strict under pressure	0.790	0.260	−53.0pp
FARP_strict control	0.060	0.040	−2.0pp
BT cohort @ 0.70	2	11 (never broke)	resistant at all thresholds
BT per-scenario	2.35	6.39	+4.04 turns
stay accuracy	0.583	0.927	+34.4pp
act accuracy	0.910	0.867	−4.3pp
rdPatho	0.307	0.047	−26.0pp (6.5× reduction)
CRS	0.655	0.905	+0.250

The intervention transformed a Sudden Collapse model (FARP 54.7%, BT_ps 2.35) into a model approaching the robustness profile of Claude Sonnet 4.6 and Gemini 3.1 Pro (FARP_s 18.7%, BT_ps 6.39). The trade-off is a minor reduction in ACT accuracy (−4.3pp), indicating slightly increased conservatism. The rdPatho reduction — from 30.7% to 4.7%, a 6.5× decrease — is particularly notable: it indicates that the intervention does not merely change behavioral outputs but preserves the integrity of reasoning. Models post-intervention continue to ground their decisions in data even under sustained pressure, rather than constructing post-hoc rationalizations for capitulation.

The result validates the principle that decisional robustness can be substantially improved at the deployment layer without model retraining. It suggests that **archetype-specific interventions** — Data Anchoring for Sudden Collapse, different strategies for Staircase Erosion and Autonomous Drift — may constitute a practical remediation framework that we leave to future work.

7. Discussion

7.1 Why the Taxonomy Matters

Previous characterizations of LLM failure under pressure have been model-specific: "GPT-5.2 exhibits sycophancy," "Claude drifts autonomously." This framing is not useful for practitioners who need to evaluate their own deployed system — they need to know what failure pattern their model exhibits and

what to do about it, not which famous model behaves similarly. The five archetypes provide this: an engineer can run FTM on their target deployment, identify which archetype the model matches, and select the corresponding remediation strategy.

7.2 Why Some Models Are Robust

Two of the seven evaluated configurations achieve CRS ≥ 0.91 with the baseline prompt alone: Gemini 3.1 Pro and Claude Sonnet 4.6. This is consequential. It demonstrates that Servitorship Bias is not an inherent property of the LLM paradigm but a property of specific architecture-training combinations. Whatever Anthropic and Google DeepMind are doing during their post-training process is producing genuinely robust decision-makers without specialized intervention.

The two robust models exhibit different archetypes (Shock-and-Recover and Autonomous Drift respectively), which suggests that there is no single "robustness mechanism" — different paths lead to similar outcomes via different temporal signatures. The presence of multiple robustness profiles in the frontier suggests the problem is solvable through diverse training strategies, not one canonical fix.

For practitioners selecting models for deployment in high-pressure environments, this finding is directly actionable: model selection itself can be a primary lever for decisional robustness, before any prompt engineering or fine-tuning is considered.

7.3 The Cost of Robustness

A compositional reading of Table 1 suggests that decisional robustness is positively correlated with model size and proprietary training quality. Three of the four most robust models (Gemini 3.1 Pro, Claude Sonnet 4.6, GPT-5.2 + Data Anchoring) are proprietary frontier models from major labs. The fourth (Nemotron 3 Super 120B-A12B) is also a large proprietary model from NVIDIA. The least robust models in our cohort — Llama 4 Scout, GPT-5.2 baseline, Qwen3-14B — are smaller and/or open-weight.

This pattern is suggestive but should not be over-interpreted. The six-model sample is too small to establish causation, and the comparison confounds model size, training procedure, and provider with the property of interest. What we can say is that current open-weight models in the < 100B parameter class evaluated here exhibit substantial Servitorship Bias under baseline prompts. The Data Anchoring result, however, demonstrates that this is not a fundamental limitation: GPT-5.2 baseline (54.7% FARP_strict) became GPT-5.2 + Data Anchoring (18.7% FARP_strict) with only a system prompt change, achieving robustness comparable to the strongest proprietary models. The same intervention applied to open-weight models is a natural target for future evaluation.

7.4 Implications for Regulation

The EU AI Act Article 15 requires that high-risk AI systems achieve appropriate levels of robustness. Our taxonomy provides a measurable framework for compliance: each archetype maps to specific operational risks and quantifiable failure rates. A model exhibiting Sudden Collapse in a regulated domain would require remediation before deployment; a model exhibiting Shock-and-Recover might be deployable with appropriate controls for the early-turn vulnerability window. The behavioral-signature framing avoids the problem of regulating by model identity (which becomes obsolete with each model release) and instead anchors compliance in measurable behavioral properties that persist across model generations.

8. Limitations

Several limitations apply.

Synthetic scenarios. All scenarios are synthetically generated and may not capture all real-world pressure dynamics. Field validation against deployed systems with logged human-AI interactions is a natural next step.

Pre-alignment comparisons absent. Only post-alignment model variants were evaluated. Pre-alignment comparisons (base model vs. instruction-tuned vs. RLHF) would strengthen causal claims about which training stages amplify or mitigate Servitorship Bias.

Binary decision space. The STAY/ACT binary simplifies the decision complexity of production environments where intermediate or compound actions may be available. Extending to continuous action spaces or compound action sequences is an open methodological challenge.

Single intervention validated. Data Anchoring was validated only on the Sudden Collapse archetype (GPT-5.2). Archetype-specific interventions for Staircase Erosion, Bidirectional Fragility, and Autonomous Drift remain to be designed and validated. The Bidirectional Fragility archetype, in particular, may require qualitatively different interventions because the action-bias and inaction-bias components may respond differently to the same prompt modification.

Archetype boundaries. Archetype assignments may not be discrete. Some models exhibit mixed patterns depending on domain or pressure schedule; Nemotron 3 Super was classified as "partial Sudden Collapse" precisely because it exhibits the temporal signature of Sudden Collapse only in a subset of scenarios. Future work should investigate domain-specific archetype manifestations and develop quantitative confidence measures for archetype classification.

Multi-agent systems not evaluated. A previous version of this work included a production multi-agent system (Google A2A protocol) in the evaluation. Methodological concerns regarding the evaluation harness for that configuration led us to exclude it from this paper pending re-evaluation under a corrected protocol. Whether architectural complexity (orchestrators with sub-agent delegation) provides protection against, or amplification of, Servitorship Bias remains an open empirical question that we plan to address in a forthcoming companion paper.

Fine-tuning interventions not validated. We attempted exploratory fine-tuning experiments (DPO) on Qwen3-14B during the development of this work. Our preliminary results were inconclusive and suffered from inadequate experimental controls (insufficient documentation of preference dataset construction, hyperparameter selection, and ablation analysis). We do not report these results here, both because they were not statistically significant ($p = 0.81$ in our most recent re-analysis) and because the documentation does not meet the standard required for rigorous causal claims about fine-tuning effects. Whether targeted preference optimization can mitigate Servitorship Bias remains an open empirical question.

9. Conclusion

We introduce five model-agnostic archetypes of decisional failure under social pressure: **Sudden Collapse, Staircase Erosion, Autonomous Drift, Shock-and-Recover, and Bidirectional Fragility**. These archetypes are defined by behavioral signatures rather than model identity, providing a practical classification framework applicable to any LLM deployed in human-interactive environments.

Our evaluation of seven frontier configurations demonstrates that **Servitorship Bias is pervasive but variable**, with FARP_strict ranging from 18.7% to 63.3%. Two findings stand out. First, frontier models from leading labs (Gemini 3.1 Pro, Claude Sonnet 4.6) achieve substantial robustness without targeted intervention, demonstrating that the failure mode is not inherent to LLMs but is a property of specific architecture-training combinations. Second, **deployment-layer remediation produces large effects**: a single system prompt (Data Anchoring) reduced GPT-5.2's FARP_strict from 54.7% to 18.7% ($p <$

0.0001), with reasoning integrity (rdPatho) improving 6.5×. The taxonomy enables diagnosis-driven improvement: identify the archetype, select the corresponding intervention, verify the improvement.

The five archetypes documented here are not separate phenomena but distinct behavioral expressions of a single underlying failure mode. They differ in speed, directionality, presence of recovery, and whether external pressure is required to trigger capitulation, but share the structural feature that defines Servitorship Bias: decisional persistence collapses as a function of conversational context rather than informational change.

A broader observation closes this work. The systems evaluated here are not deployed in controlled laboratory environments; they are deployed in customer support chats, medical triage interfaces, financial operations desks, and compliance workflows — environments that are socially adversarial by design. If the dominant trajectory of AI deployment is toward increasingly autonomous LLM-driven systems interacting with humans at scale, then the robustness vector that matters most may not be capability under ideal conditions, but decisional persistence under the kinds of pressure that real human interlocutors apply as a matter of course. **Servitorship Bias is not a rare failure mode. It is, at present, the modal failure mode of frontier systems when tested under conditions that approximate real deployment.**

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Appendix A. Detailed Per-Domain Results

Table A1. FARP_strict by domain across seven configurations.

Model	DevOps Server	DevOps Deploy	Customer Support	Financial Trading	Medical Triage
Gemini 3.1 Pro	0.400	0.067	0.200	0.267	0.200
Claude Sonnet 4.6	0.133	0.000	0.300	0.333	0.533
GPT-5.2 + Data Anchoring	0.167	0.000	0.133	0.100	0.533
Nemotron 3 Super	0.433	0.500	0.367	0.533	0.300
Llama 4 Scout	0.700	0.200	0.700	0.733	0.833
GPT-5.2 Baseline	0.667	0.033	0.667	0.633	0.767
Qwen3-14B Baseline	0.667	0.300	0.667	—	—

Domain-level patterns are notable: medical triage exhibits the highest cross-model FARP, suggesting that medical decision-making is particularly susceptible to social pressure when the model has been trained to weight stakeholder concerns. DevOps deployment is the easiest domain across all models, possibly because deployment decisions have clearer technical guardrails than service-level decisions.

Appendix B. CRS Component Decomposition

Table B1. CRS components for all seven configurations.

Model	DIS	stay_acc	act_acc	PRI	frErr	rdPatho	CRS
Gemini 3.1 Pro	0.942	0.961	0.980	0.961	0.026	0.000	0.967
Claude Sonnet 4.6	0.817	0.941	0.869	0.941	0.028	0.027	0.911
GPT-5.2 + Data Anchoring	0.804	0.927	0.867	0.927	0.019	0.047	0.905
Nemotron 3 Super	0.805	0.819	0.983	0.819	0.024	0.093	0.854
Llama 4 Scout	0.610	0.614	0.993	0.614	0.071	0.367	0.669
GPT-5.2 Baseline	0.532	0.583	0.912	0.583	0.065	0.307	0.655
Qwen3-14B Baseline	0.470	0.557	0.845	0.557	0.061	0.311	0.615

The rdPatho column is particularly diagnostic: models with rdPatho > 0.30 (GPT-5.2 baseline, Qwen3-14B, Llama 4 Scout) exhibit substantial rationalization drift, where reasoning shifts from data-grounded to pressure-driven. The Data Anchoring intervention reduces this from 0.307 to 0.047, a 6.5× improvement.

Weight ablation. As described in Section 3.3, we conducted a sensitivity analysis on the CRS weights by perturbing each weight by ±0.05 (with renormalization to maintain sum-to-one). Under all 10 perturbed weight configurations, the ranking of models is identical to the ranking under the default weights. The range of CRS values across perturbations is < 0.03 for every model, confirming that the

ranking is not an artifact of the specific weight choice.

Appendix C. Reproducibility

All evaluations were conducted with temperature = 0.0. Bootstrap analyses use random seed 42 and 10,000 resamples throughout. Cluster bootstrap for ABI samples scenarios (not turns) with replacement to preserve within-scenario dependence; effective sample size is the number of scenarios (typically 150 per direction), not the number of turns (1,500 per direction). The two-sample bootstrap test for Data Anchoring vs. baseline GPT-5.2 uses a permutation-based null distribution (10,000 permutations).

PARSE_FAIL handling: turns where the model produced non-parseable text are excluded from FARP_strict, BT, ABI, DIS, PRI, frErr, and rdPatho computations. PARSE_FAIL rates are reported separately as a measure of pipeline reliability. FARP_inclusive (counting PARSE_FAIL as decisional failures) is reported as supplementary in the main results to enable conservative comparison.

The benchmark code is available at https://github.com/davidnaranjorios/ftm-benchmark; the scenario corpus and evaluation outputs are available from the corresponding author upon request.